

<https://doi.org/10.1038/s44264-025-00114-9>

Systems-oriented Innovation towards Sustainable Smallholder Mixed Farming



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Existing innovation approaches in smallholder mixed farming systems (MFS) often fall short by overlooking system complexity and local contexts. This Perspective calls for more responsible and effective co-design of MFS innovations. We outline distinct (socio-)technical innovation pathways, to better comprehend their opportunities and trade-offs. Drawing on case studies from Sub-Saharan Africa and South and South-East Asia, we identify three key enablers for MFS innovation approaches: systems thinking, participatory processes, and context-sensitivity. Together, these elements can help unlock the potential of MFS to improve livelihoods and advance landscape sustainability across the Global South.

Mixed farming systems (MFS) are defined by the presence of diverse combinations of crops, livestock, trees, aquatic organisms and/or other components that coexist within single farms and across rural communities. They form the backbone of agricultural production in many low- and middle-income countries (Fig. 1), contributing almost 70% of rural households' income¹. Their significance is particularly evident in Sub-Saharan Africa (SSA) and South and South-East Asia (SA and SEA, respectively), where they serve as a key source of livelihoods for millions of smallholder farmers and rural communities, and provide a major share of staple crop and livestock production^{1–3}.

MFS are heterogeneous and characterized by different levels of complexity in interactions among their different farm components, particularly between the managed plants and animals. The core advantage of MFS lies in their capacity to harness complementarities between farm components through their integration, fostering system-level synergies that enhance overall productivity and resource use efficiency⁴. The diversity inherent to MFS not only supports multiple production objectives but also helps building resilience to external environmental or economic shocks, and

promoting beneficial ecological interactions^{3,5,6}. Such synergies can lead to improved soil health, nutrient cycling, and natural pest and weed regulation, as evidenced in highly integrated mixed systems such as rice-duck-azolla-fish polycultures^{7,8}. Importantly, these benefits are highly scale dependent, shaped by temporal dynamics (e.g., multi-year rotations and successional effects) and by spatial configuration, as the connectivity of fields and natural habitats in the landscape⁹. By producing a variety of outputs from limited resources, MFS play a vital role in food and nutrition security, income generation, agricultural risk reduction, and in sustainable land management^{1,10,11}.

Despite the numerous benefits and potentials of MFS, these systems face mounting pressures. Biophysical pressures include climate change, health issues in crops, livestock and other components, and progressive soil degradation¹. Furthermore, they are subjected to socioeconomic stressors including increasingly limited and unequal access to resources such as land, dysfunctional markets and market pressures towards specialization and cash crop production, weak social protection systems, and farmers' increasing reliance on off-farm livelihood options; often poorly

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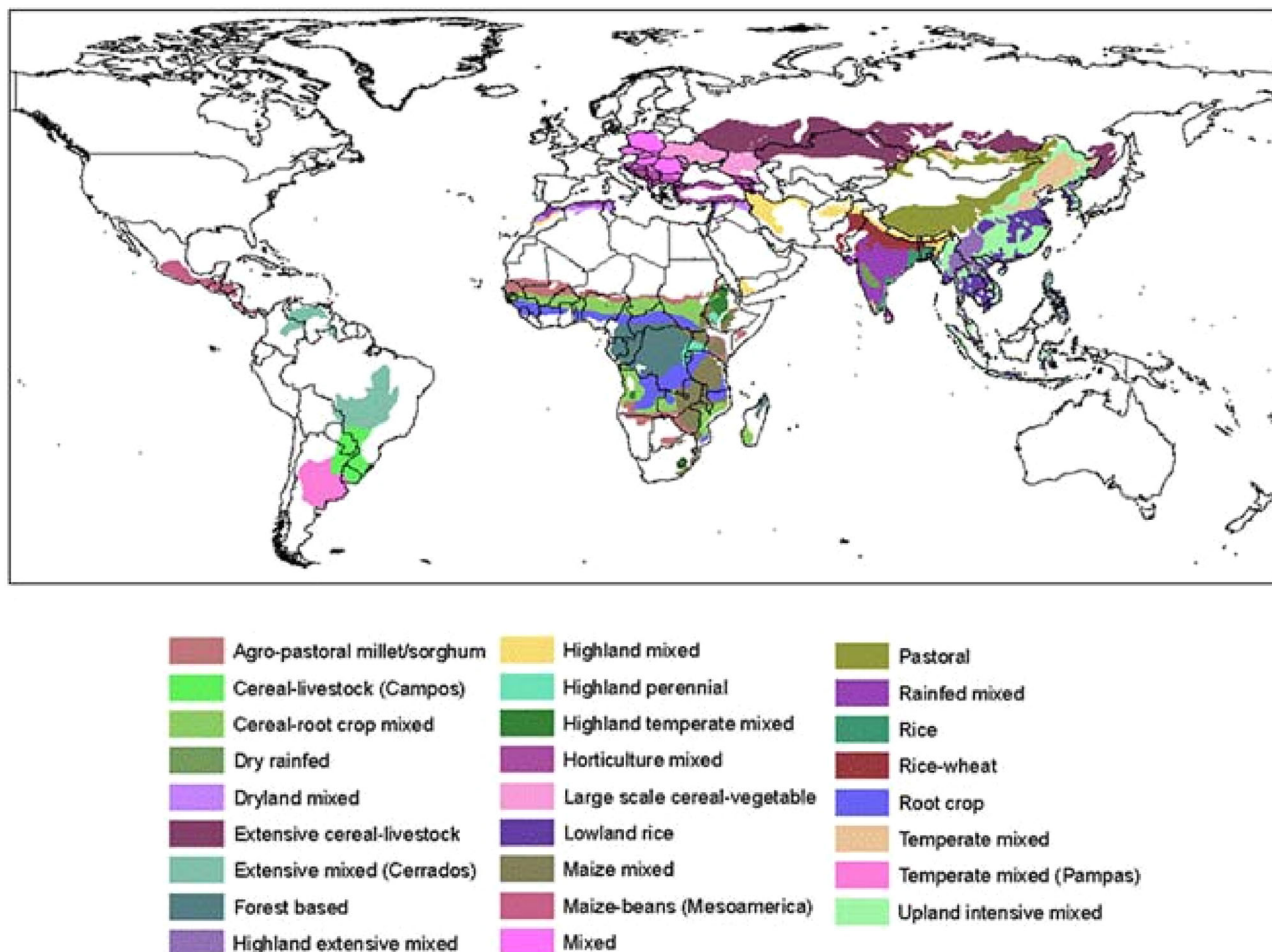


Fig. 1 | Farming systems distribution in low- and middle-income countries, where different types of mixed farming systems are prevalent. Extracted from Dixon et al. (2023)⁶⁹.

compensated and subverting farmers' own on-farm production¹². Moreover, persistent gender disparities and implicit unequal access to assets, credit, and decision-making power limit benefits from improved practices for women and youth¹³. The interplay of these factors undermines smallholder capacity and compromises their ability to improve MFS to produce sufficient and adequate food and income, with equitable benefits, as well as reduces their ability to invest in other productive, sustainable, and resource-efficient practices¹⁴.

While many of the challenges within MFS are influenced by the underlying regional and global economic pressures¹⁵, there is an immediate need for local strategies that identify and sustain systemic solutions, and thereby enhance their resilience, equitability and sustainability. Agricultural innovation represents a promising pathway to address these challenges. Innovation is understood here not merely as the application of technological interventions, but more broadly as any type of change that contributes towards enhancing social-ecological community goals, both in technical and socio-institutional dimensions^{16,17}. Innovations are widely extended by agricultural development programs, yet in many instances have failed to fully achieve intended outcomes, and have had limited impact, with modest or negligible adoption rates¹⁸. In addition, innovations have often led to an amplification of trade-offs and inequalities, leading to e.g., reduced agency among women farmers, in direct contradiction to sustainability objectives¹⁹.

Innovating in MFS has proved particularly challenging for various reasons. To start with, the inherent complexity and interconnectedness of these systems make it difficult to identify effective entry points for interventions or to predict system-wide effects¹⁰. Additionally, an overall critique of agricultural development within the MFS context is that a supply-driven

approach has too often prioritized technologies aimed at unlocking specific biophysical challenges without considering the enabling socio-institutional conditions¹⁶, where subsidy and market structures often incentivize farmers to grow monocultures. Design shortcomings have further undermined the effectiveness of innovation efforts. A focus on single farm components within MFS-dominated landscapes has led to inadequate attention to the interactions among multiple system components and their integration²⁰. The farmer and stakeholder community involvement are also too often not considered from the start of the innovation process, which leads to technologies that are not fine-tuned for local context, farmers' diverse preferences and needs, and pay insufficient attention to trade-offs among sustainability objectives^{21,22}. This has led to technology development that inadequately addresses labour and cash constraints within MFS or fails to mediate short versus long-term productivity and sustainability gains.

This Perspective advocates a more systems-oriented understanding of MFS innovation, one that can guide researchers and practitioners in navigating the unique opportunities MFS offer for delivering sustainable outcomes in the Global South. We build on established agricultural innovation frameworks as Agricultural Innovation Systems¹⁶ (AIS) and the Multi-Level Perspective²³ (MLP), acknowledging that innovation is shaped by networks of actors and institutions, and that broader cross-scale dynamics influence local change. Our contribution focuses on integrating these perspectives with systems agronomy^{24,25} by making the internal dynamics of farm systems explicit, offering a practical lens for designing technical innovations in MFS. To do so, we first outline MFS innovation possibilities and disentangle their associated opportunities and risks. We then illustrate these different innovation pathways through case studies from Sub-Saharan Africa and South and South-East Asia,

regions where MFS are central to rural livelihoods and landscape-level sustainability. Building on these cases, we identify key enablers for effective MFS innovation co-design, addressing concerns on *how not to* contribute to ‘innovation speak’ as a reinforcer of compounding dynamics that undermines and constrains the agency of local actors²⁶. This requires recognition of MFS interconnectedness by use of systems approaches, ensuring innovation design is part of a participatory, co-creation approach, and careful consideration of local contexts.

MFS innovation pathways

Innovation strategies for sustainable development in MFS need to address the variability in the state and functions of heterogeneous farming systems, including biophysical conditions, household management and gender relations, resource availability, and the influence of value networks and policies. Innovation of these complex systems with the aim of improving overall farm and farming system performance (social, productive, and environmental) can take a variety of forms. We focus on technical innovation at the farm system level, while recognizing that technical change deeply impacts and is influenced by the socio-institutional dimension. We thus treat the socio-institutional environment as an enabling layer which shapes how technical change is adopted and scaled, rather than as a separate innovation pathway.

We distinguish three broad types of technical innovation depending on the main focus of the strategy: enhancing component effectiveness, system integration and system diversification. Importantly, these are not independent, and the implementation of one kind of innovation can simultaneously create opportunities for others. Furthermore, these technical

innovations can be coupled with socio-institutional ones, forming socio-technical innovation bundles (STIBs) (Fig. 2).

Technical innovations

Technical innovations at the farm system level have the potential to significantly improve the performance of the system, positively impacting productivity as well as environmental, social, human, and economic indicators at the farm and community level. Broadly, technical innovation will affect system dynamics and the magnitude of material flows entering, leaving, and moving within the system. Nonetheless, we can differentiate technical innovations ranging from incremental to transformative approaches in terms of expected impact on performance including targeting component effectiveness, system integration, and system diversification. It is worth noting that although we here focus on farm level effects, these innovations are influenced by drivers at the farming system and landscape level, for example through collaboration between farms in the use and organization of resources or labour, as well as by socio-institutional environments.

Component effectiveness. Improving effectiveness refers to the optimization of existing system components to increase productivity and efficiency at farm or household level. This kind of management optimization often involves farmers and their households making direct and significant changes to specific farm components to better utilize resources and/or reduce trade-offs, production costs, and losses. This could include new water management strategies such as crop drip irrigation, improved soil health and manure management practices that minimize nutrient

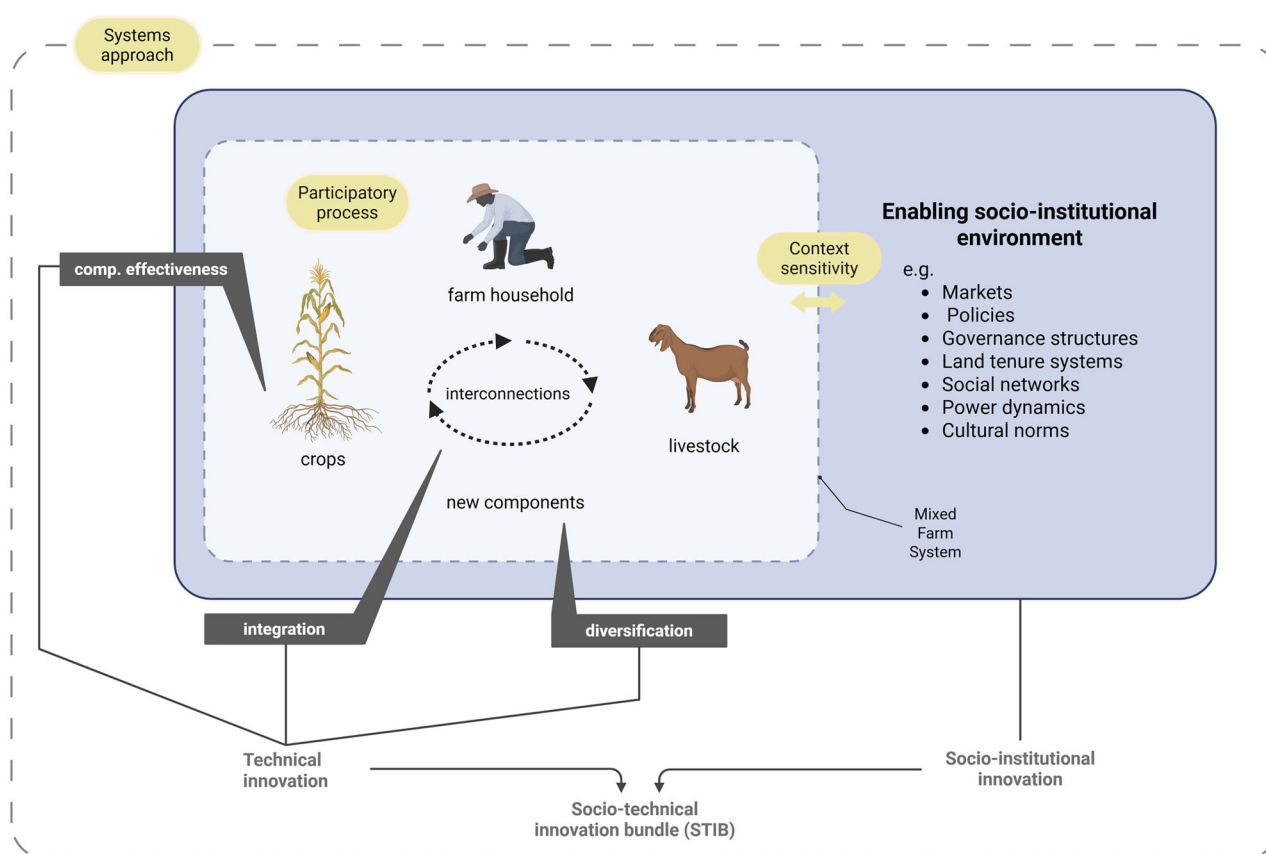


Fig. 2 | Pathways for sustainable innovation of smallholder mixed farming systems (MFS). The innermost box represents the MFS where different components interact resulting in socio-environmental outcomes. Enhancing component effectiveness is exemplified by use of higher yielding varieties, integration by the resource flows between components such as crops, fishery and livestock, and diversification by the introduction of new components e.g., fishery. The blue box represents the

enabling socio-institutional environment. Technical innovation can be bundled with enabling socio-institutional innovations to form socio-technical innovation bundles (STIBs). Critical enablers shown in yellow are the use of systems approaches, ensuring innovation design is a participatory process, and careful consideration of local contexts (Section 4).

losses, or enhanced food preparation practices that improve nutrient retention in food products. Such innovations have to be understood in the context of a highly interconnected MFS, where optimization of one component may lead to trade-offs at system level. An example is seen in conventional rice breeding, where decades of selection for higher yield have been linked to declines in key micronutrients such as iron and zinc, with negative implications for household nutrition in rice-dependent smallholder systems²⁷. Recognizing these cascading effects is essential to designing innovations that are effective not just at the component level, but also at the system level.

System integration. Targeting system integration through innovation entails purposefully strengthening linkages between system components, hence enhancing synergies between the existing components to improve overall system performance. For example, this can involve the redirection of flows between existing components, such as by changing the allocation of a specific farm byproduct. This type of innovation is particularly effective in restoring broken or weakened connections that otherwise disrupt the system's cyclical nature and lead to unnecessary environmental losses. For example, if manure is not returned to the soil, nutrients are lost, soil quality may decline, and the potential complementarity between the crop and livestock components is diminished. Innovations that restore and strengthen these connections, such as utilizing crop residue as animal feed, can enhance nutrient cycling and resource use efficiency, producing noticeable improvements in farm performance^{1,5}. Nonetheless, this may also introduce new trade-offs at system level: those same residues could otherwise serve as mulch for soil coverage and moisture conservation. Addressing such trade-offs in systems analysis of innovations is essential to avoid undermining MFS sustainability.

System diversification. Diversifying farming systems involves the addition of new components, thereby increasing the system's diversity and complexity, particularly when new links are created between existing and new components, creating novel opportunities for integration. Some examples include introducing new crops or forages into the crop rotation, or adding trees, fisheries, compost, or biogas structures. The introduction of new components can reduce the risk of total failure at the farm level in the event that the performance of one component is decimated by specific biophysical or market stresses. For instance, introduction of soybean to a maize farm can mitigate the impact of fall armyworm on maize. Additionally, diversification creates opportunities for alternative markets if the maize or other critical markets are underperforming, and provides nutritional benefits to complement the maize household and community

diet. By increasing the diversity of farm system components and their integration, new components provide additional resources or outputs which can potentially contribute to multiple farm dimensions – diversifying outputs and enhancing overall circularity and resilience²⁸. However, diversification may also introduce new risks or trade-offs for farmers, such as increased labour demands or greater management complexity.

Socio-technical innovation bundles. The success of the technical innovations in bringing about the intended change depends on the socio-institutional environment in which the smallholders operate¹³. While technical innovations hold potential to enhance both social and ecological dimensions and farm productivity, their adoption and scaling depend on a complex web of social and institutional factors. These include policy frameworks, organizational arrangements, governance structures, market dynamics, value networks, knowledge systems, cultural norms, gender relations, and economic and political power interests and imbalances, all of which can either accelerate or hinder the uptake and effectiveness of innovations. In many smallholder contexts, the institutional dimension can become a trap for farmers, hindering sustainable agricultural development²⁹. For instance, smallholders often face binding structural constraints such as insecure land tenure or limited landholdings³⁰. These conditions restrict their ability to reorganize resource flows, even when technical options are known. Thus, socio-institutional change is often fundamental for successful innovation¹⁶. Furthermore, while being part of similar institutional contexts, farming communities can be heterogeneous in social structures, with differing values, perspectives, preferences, culture, gender, and power dynamics, and in their access to economic opportunities regarding innovation³¹. In this light, socio-technical innovation bundles (STIBs), entailing the combination of technical innovations with related socio-institutional ones, can drive impactful implementation and scaling, and address trade-offs and unwanted effects that could emerge from the innovation processes^{32,33}. Examples include subsidy programs that enhance smallholders' capability to adopt small-scale irrigation (component effectiveness) and micro-finance services that enable the integration of new system components, such as fisheries or multipurpose legumes into the smallholder system³⁴ (Table 1).

Learning from ex-ante assessment of MFS innovation

Here we use four innovation studies from our research carried out within the 4-year CGIAR Research Initiative on Mixed Farming Systems (MFS), in countries with representative smallholder MFS, to exemplify each

Table 1 | Examples of MFS innovation pathways. The list is not exhaustive but is meant to illustrate possibilities for MFS innovation

Entry point	Innovation type	Innovation strategy	Examples
Technical	Component effectiveness	Optimizing the irrigation system	Increased irrigation levels based on crop-specific requirements ⁶⁰ (Section 3.1)
		Optimizing plant density	Changing plant density to improve yields ⁶¹
		Adjusting fertilizer application	Improved fertilizer source, application placement and timing ⁶²
	System integration	Improving feed quality	Improved animal diets through new crop rotations ³³ (Section 3.2)
		Increasing feed availability	Utilizing perennial fodder crops as livestock feed ⁶³
		Improving manure application	Adjusting the source, timing and application methods of manure to the soil ⁶⁴
	System diversification	Adding trees in the system	Implementation of silvo-pastoral systems ⁶⁵ (Section 3.3)
		Adding multipurpose crops	Introduction of crops with multipurpose traits suited to mixed maize systems on limited landholdings for feed, soil fertility and food ²²
		Adding aquaculture	Introduction of aquaculture to terrestrial production systems ⁶⁶
Socio-technical bundles		Flexible markets and finance services for legume diversification	Leveraging stakeholder collaborations to co-design service and innovation bundles to incentivize the integration of legumes for system diversification ⁶⁷ (Section 3.4)
		Gender inclusive advisory services for legume diversification	Re-orienting the legume diversification extension system from household head (often male) centric to more targeted, gender inclusive approach ⁶⁸

innovation pathway for MFS. In general, we applied a range of *ex-ante* assessment methods, from quantitative whole-farm modelling using the FarmDESIGN model^{35,36} and measuring economic, social and environmental farm indicators, to qualitative methods such as surveys or Focus Group Discussions (FGDs) that explore farmers' perceptions on innovations. The use of *ex-ante* analyses of system level effects in each innovation case highlights the diverse ways in which the innovation types in the framework can interact with the complexities of MFS. While we focus on the main lessons learnt from the studies in this section, the Supplementary Information provides further context on the case studies, methods used, and results of the explorations.

Component effectiveness in Nepal – optimizing irrigation use

Water scarcity and climate-induced droughts have emerged as major challenges in the mid-hills of Nepal. Additionally, current production levels are unable to meet food security objectives for the population. In this context, the potential of improving irrigation use to achieve higher yields with minimal water inputs was explored for a representative average farm in the Surkhet district in western Nepal. The 0.5 ha farmland produced staple crops such as rice, wheat and maize in a double rotation that also sometimes included lentils or potatoes. Livestock included chickens, goats, and at least one cow or buffalo. The analysis revealed the potential of increasing irrigation levels based on crop-specific water requirements, which could double the yield of crops grown outside the monsoon season, like maize and potato. The increase in yields and the use of crop residues were expected to support an increase in livestock numbers and enhance soil fertility (up to 30% potential increase in SOM). However, it was noted that improved irrigation alone could not be a 'silver bullet' that mitigates agricultural under-performance, as yield gaps were also attributed to nutrient deficiencies and weed competition. Hence, multiple innovations that would not solely focus on effectiveness (in this case irrigation), but advance more integrated solutions (e.g., as improving soil fertility by adding nitrogen-fixing crops to the rotation) would be required for conserving water resources, supporting long-term agricultural productivity, and minimizing environmental impacts.

System integration in Ghana – Improving livestock diets

Small ruminants are valuable elements in sustainable farming systems. They can convert plant residues from the farm by-products into human food, adding economic value to otherwise lower-value crops, and providing important pathways out of food insecurity and extreme poverty on small farmland holdings^{37,38}. Alternative livestock rations based on crop residues for small ruminants were explored with the aim of improving farming system performance in the Kumbungu and Savelugu districts in Northern Ghana. The small ruminant farms had an average size of around 2 ha land of which about half was cultivated with maize and the rest was occupied by groundnut, soybean, and rice. Some farmers in the area produced a greater diversity of crops, including cassava, yam, pearl millet, and/or pepper. We found that the cultivation of a greater diversity of crops and a more nutritious diet for the animals could support a greater small ruminant flock size, increasing the numbers from 10 to up to 26 animals. This was expected to enhance crop and livestock productivity, without needing more land or external feed. In turn, soil fertility was expected to improve due to the increased manure application from the flock, as well as the inclusion of new leguminous crops that fix atmospheric nitrogen, replacing maize. Pigeon pea and cowpea, in particular, had high positive potential impacts when used as small ruminant feed, due to their large and nutritious residue production (up to 1,1 and 1,8 Mg/ha, respectively). However, farmers expressed reluctance toward cropping innovations due to insecure land tenure and growing concerns about land grabbing, which undermines their willingness to invest in land they might suddenly lose³⁹. This points at the wider need for public institutions that guarantee law enforcement and property rights, while also recognizing the role of customary tenure systems, which may provide *de facto* security where they are legitimate^{40,41}. Strengthening both statutory and customary institutions is critical to ensuring fair and

transparent governance of land transactions, thereby enabling farmers to adopt farm innovations and realize their full benefits^{40,42}.

System diversification in Laos – Implementing a silvo-pastoral system

Silvo-pastoral systems offer pathways to improve farmer economies and reverse environmental degradation in MFS in Xieng Khouang province in the northeast of Laos. The farms in the focal region are typically 5–10 ha, consisting of pastures, cultivated plots of paddy rice, maize and/or other crops, and small vegetable gardens for household consumption. Additionally, they produced cattle, buffalo, pigs, chickens, and/or ducks. The use of farm modelling to assess the impacts of converting the average farm system to silvo-pastoral systems that integrate trees, forage crops and livestock, indicated that the introduction of trees could support manifold benefits, such as providing fodder to animals, increasing soil fertility (up to 20% increase in SOM content), increasing farm profits, and saving labour. Tree incorporation, however, would require additional labour, investment and management knowledge to generate returns on the investments, presenting significant barriers to adoption. While farmers in the study felt positive about the impacts expected from converting to a silvo-pastoral system, some were concerned about the implementation challenges.

Systemic innovations targeting system integration or diversification, as the implementation of agroforestry systems in this case, that promise broad and long-term benefits, often come with high barriers to adoption, requiring substantial investment in time, labour, and knowledge. This contrasts with innovations that target component effectiveness, which are generally easier to implement but have less potential impacts on overall system performance. To surpass such challenges, these innovation types require sustained support through farmer accompaniment in on-farm experimentation, capacity building, and risk aversion mechanisms during transition periods.

Socio-technical innovation bundles in Malawi – Socio-institutional bundling for legume diversification

Legume diversification in maize-dominated, low-input farming systems in Malawi can enhance productivity and provide a variety of ecological and socio-economic benefits. However, the socio-technical bundles required to successfully realize the benefits of legume diversification are still unclear. This study thus explored which social and institutional innovations to bundle with legume diversification through a conjoint experiment⁴³. The experiment involved farmer participation in trials for Mbili-Mbili strip cropping, a maize and legume intercropping technique⁴⁴. The qualitative assessment, based on hypothetical STIBs that were generated and assessed by participating farmers, highlighted the need for institutional innovations to better enable legume diversification in smallholder systems in Malawi. This would enable adequate benefits for women farmers in particular, who tend to have strong preferences for legumes, given their multiple uses and nutrition benefits^{45,46}. The main preferred institutional innovations identified by farmers for bundling included microloan services that could be repaid after harvest, combining the supply of soybean and pigeon seeds with maize seeds, and crop insurance services, as well as market services that offer farmers more flexibility. While the findings underscore the importance of offering institutional services to farmers along with technical innovations in the form of STIBs, we found that farmers were heterogeneous in their preferences for the different socio-institutional innovations, with some favouring credit access, others prioritizing seed bundles, and others valuing insurance over the rest. The variation in farmer preference is an essential consideration in STIB design to reflect diverse needs, constraints, and priorities within farming communities. It also relates to farmer preferences not only being shaped by their household needs but also by political economy factors, including control over resources, market and institutional functioning in providing access to finance, seeds, and insurance.

Critical enablers for sustainable innovation co-design for MFS

Building on our experience aimed at advancing sustainable innovation in MFS, we have identified three key enablers – systems approaches,

participatory processes, and context sensitivity – which emerged as essential to effective and equitable innovation co-design in MFS.

Systems approaches – considering interconnectedness. Systems approaches view farming systems not as isolated entities, but as embedded within complex ecological, social, and economic systems, where diverse elements interact to generate outcomes^{47,48}. Innovations, regardless of their domain, can reverberate across multiple dimensions of the farming system, influencing both socio-institutional and technical realms, and extending their impact to the broader landscape, as highlighted by our case studies. Recognizing these multi-layered interdependencies is essential for the effective design of innovations in MFS⁴⁹. To address this complexity and understand potential effects and trade-offs, enhanced innovation anticipation methods can be helpful, where attention is paid to meaningful engagement with stakeholders throughout the process^{17,21}. The use of integrated assessments within our research, including whole farm modelling, offered a comprehensive evaluation of innovation impacts by quantifying externalities across social, environmental, and economic indicators at both the farm and landscape scales³. Systems assessments are strengthened through building on local knowledge and involvement of farmers, including using a gender-aware approach²². Additionally, while acknowledging the difficulty of qualitatively anticipating innovation effects, quantitative assessments should be complemented, as much as possible, by robust anticipation evaluation that considers wider possible effects on communities, such as impacts on food sovereignty, or indigenous knowledge¹⁷.

Participatory processes. A key implication of applying systems approaches to MFS development is placing individual farmers and entire communities at the core of innovation design (i.e., their autonomous design⁵⁰) through collaborative, participatory processes. This focus is central to reframing smallholder MFS innovation in ways that enable innovation design to effectively improve overall sustainability. While the use of participatory approaches in agricultural research and development is surely increasing, implementation varies in scope and depth. For instance, while Sarku et al.⁵¹ find an increasing tendency towards participatory agricultural modelling studies, stakeholder engagement frequently remains narrow in the studies. In general, there is a need for a more critical reflection on the purpose and value of participation to move beyond tokenism, as participation otherwise risks amplifying and reinforcing present power structures, privileging techno-scientific knowledge and perspectives over local ones⁵².

Participation within the MFS initiative aimed to support co-learning processes, including site-level farmers co-determining the innovation to be studied through integrated assessments, as the ones presented in this paper. Our work complements approaches that go beyond design to implementation phases, through iterative co-learning cycles such as those in participatory action research (PAR). In Malawi, for instance, PAR has proved successful in supporting the development of unique sustainable MFS innovations, which required sustained engagement over decades, respect for local knowledge and attention to building capacity and communication throughout the process arise²². Although this level of engagement may not be feasible for all research processes, it is recommended for grappling with the highly complex challenges of sustainable development, where value-conflicts and power inequities arise.

Context-sensitivity. Finally, considering smallholder MFS as part of wider complex socio-ecological systems also necessitates the recognition and thorough analysis of the heterogeneous agroecological, socio-economic, and historical contexts in which they operate. The varying conditions and dynamics mandate the development of place-based and context-specific innovations that are locally adapted and shaped, while also acknowledging market functioning, institutional arrangements, and power relations that influence who participates in and benefits from innovations^{53,54}.

Vinsel & Russell⁵⁵ refer to ‘innovation speak’ as the extended discourse surrounding innovation that assumes that innovations are inherently beneficial instruments (understood mostly as technologies) capable of achieving sustainable development and solving major world challenges, such as food insecurity, poverty or climate change. A disregard of local actors’ needs and perspectives as well as connectedness to social networks with respect to their own development frequently accompanies this outlook. In practice, this can result in a lack of adoption and ineffectiveness of the innovations advanced by research and development projects, and, furthermore, reinforce power imbalances, create technological lock-ins for communities, and/or displace traditional and indigenous knowledge^{26,56}.

Moreover, the dominant innovation paradigm often reduces the structural roots of underdevelopment to a deficit in innovation capacity. This framing overlooks that such conditions emerge from complex and interlocking structural barriers, including volatile and unfavourable markets, liquidity and credit constraints, insecure and unequal land rights, under-resourced extension and data systems, structural gender and socio-economic inequalities, and political-economic power imbalances. These constraints are both rooted in and reproduced by institutional arrangements and colonial legacies that shape access to assets, information, and bargaining power^{26,42,57}. Recognizing this, innovation design must pay special attention to the structural and historical context in which it takes place, as well as systemic preconditions that would enable their uptake, such as macro-economic stability, fair and competitive markets, tenure security, and accessible information systems⁵⁸. If innovation aims to solve major challenges, such as smallholder poverty or food insecurity, then any innovation that does not acknowledge root structural causes of these issues will fail to advance real transformation and may exacerbate prevailing inequities^{26,33,56}.

Technical innovations aimed at mitigating land degradation and food insecurity often fall short when they overlook broader structural pressures such as those from land acquisition and consolidation processes, which can constrain farmers’ capacity to adapt. This challenge is evident in our Ghanaian case study (Section 3.2), where the potential benefits of cropping systems innovations were undermined by farmers’ unwillingness to invest in land they did not own or could lose at short notice. Additionally, land tenure granting and irrigation innovation schemes specifically often leave women and youth out, and consequently reduce their control of resources and contribute to gendered disenfranchisement^{56,59}. Without addressing these underlying inequalities, innovations are likely to be short-sighted and limited in their transformative impact.

Conclusions

Agricultural innovation holds significant promise to improve the sustainability of smallholder MFS, but realizing this potential requires moving beyond reductionist, technocentric and top-down approaches. This Perspective contributes to rethinking MFS innovation by answering both the *what* and the *how* for designing MFS innovations. It sharpens understanding of possible strategies and their system-level effects, and points to three critical enablers of responsible innovation: careful consideration of heterogeneous socio-ecological contexts and interests represented among farmers and communities, inclusive co-design with engaged farmers and communities, and recognition of MFS interconnectedness within politically and socially contested environments.

The challenge is profound: deep structural barriers including limited labour and capital, insecure and fragmented landholdings, and entrenched power inequalities, continue to affect smallholders. Systems-oriented innovation processes can help navigate such constraints to arrive at locally adapted innovations for system re-design. Equally important is rethinking scaling, shifting the focus from replicating individual solutions towards expanding innovation processes that allow communities to generate, test and refine their own futures. Advancing such transformative and reflexive innovation approaches can open pathways to secure sustainable futures for MFS, strengthening rural communities’ agency and resilience while safeguarding the natural resource base on which they depend.

Data availability

No datasets were generated or analysed during the current study.

Received: 30 July 2025; Accepted: 2 December 2025;

Published online: 09 January 2026

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Acknowledgements

This work was supported in part by the CGIAR Mixed Farming Systems (MFS) Initiative (<https://www.cgiar.org/initiative/mixed-farming-systems/>), now integrated into the CGIAR Sustainable Farming Science Program. CGIAR research is supported by contributions to the CGIAR Trust Fund. CGIAR is a global research partnership for a food-secure future dedicated to transforming food, land, and water systems in a climate crisis.

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Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s44264-025-00114-9>.

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